

# ThaiSLM — Result Report v6

**Scope:** v6 = v5 + the new th-sl.com dictionary (7,719 clips), one extractor for all video (TTRS re-extracted at 25 fps), a standard data-prep layer on S3 ( data\_prep/ ), and the improvements asked for in prompt\_7 / prompt\_8 — each one tried, measured on held-out data, and **kept only if it helped**. Architecture: architecture\_v6.md . Representation space: latent\_space\_v6.md . Data schema: cloud\_s3/data\_prep/SCHEMA.md . Entry point: python main.py ...; post-training check: lab.ipynb .

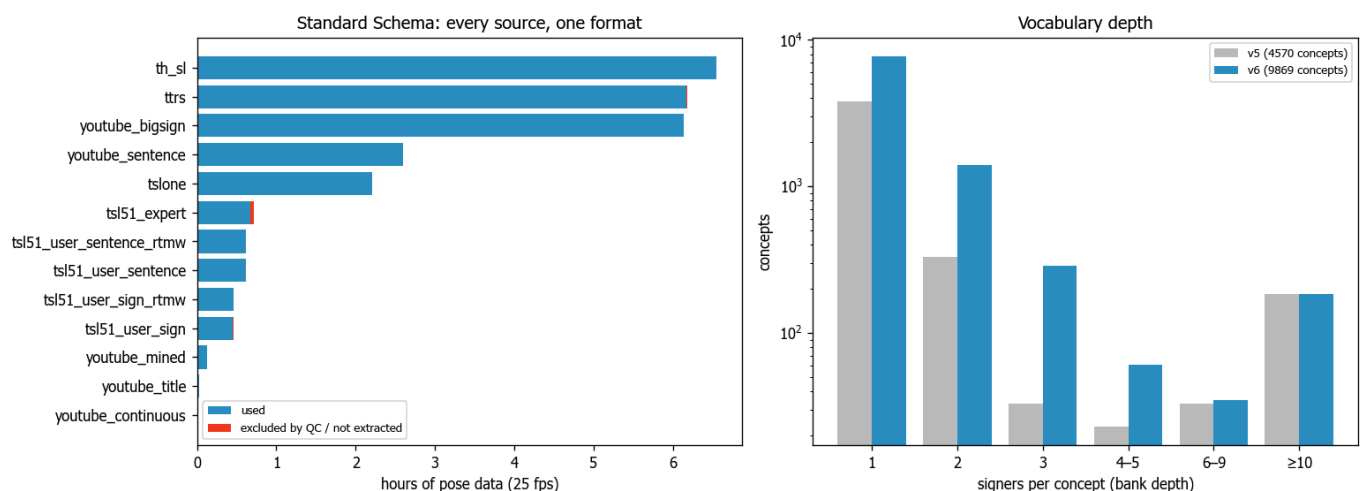
## 0. TL;DR

| question                                                                   | answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is v6 better than v5 on real-world video (data_test, from the raw videos)? | <b>Yes.</b> Reference word top-1 in some segment <b>9/22</b> (v5 7/22) · in top-5 <b>14/22</b> (v5 10/22) · accepted words correct <b>7/7 = 100 %</b> (v5 6/12 = 50 %). Newly found: ร้องไห้, ต้ม (top-1) and น้ำ (#2) — words that had 1–3 examples in v5 and got th_sl signers. Word-count error per video 1.0 (v5 0.8).                                                                                                                                                                                                     |
| Better on held-out signers (isolated signs)?                               | <b>Yes, on the same queries and the same bank</b> (v5 encoder re-scored on v6 data): TTRS new signers R@1 0.155 → <b>0.190</b> , TSL-ONE-S 5 new signers 0.864 → <b>0.890</b> , th_sl unseen signer 0.090 → <b>0.105</b> , YouTube signers 0.176 → <b>0.235</b> . Null-vs-sign AUC (RTMW) 0.86 → <b>0.95</b> . TSL51 researcher (1 signer, 51 words) 0.368 → 0.342 (RTMW) / 0.633 → 0.620 (MediaPipe) — slightly lower (§4).                                                                                                   |
| Better on continuous signing (held-out TSL51 templates)?                   | <b>More trustworthy and much better on fast signing; slightly worse at natural pace.</b> Accepted-tier precision <b>0.85</b> (RTMW, v5 0.76) / <b>0.91</b> (MediaPipe, v5 0.84). At 2× speed: exact word count <b>67 %</b> (v5 27 %), recall@1 <b>0.456</b> (v5 0.382), WER <b>0.558</b> (v5 0.597). At 1×: recall@1 0.487 (v5 0.473) but segment F1 0.793 (v5 0.835), exact count 64 % (v5 71 %), WER 0.571 (v5 0.524 before / 0.467 after the grammar re-rank). §5 explains why (the vocabulary doubled) and what was tried. |
| Did the new data help?                                                     | <b>Yes, measured by training without it</b> (same bank): without th_sl, TTRS new signers 0.131 (with: 0.190), TSL-ONE-S 0.858 (0.890). Without the MediaPipe sources, TSL-ONE-S 0.780 — neither source is noise. Vocabulary 4,570 → <b>9,869 concepts</b> ; concepts with ≥ 2 signers 605 → <b>1,969</b> .                                                                                                                                                                                                                     |
| Is the data clean / one standard?                                          | One Standard Schema for 19,897 clips / 26.7 h (22.7 h RTMW = the real-video extractor), every file and column documented in data_prep/SCHEMA.md . QC removed only 58 clips (45 duplicates of th_sl videos inside TSL51, 2 body-part close-ups with no signer, 1 corrupt video ...); a first QC rule that would have dropped 175 good TSL-ONE-S clips was caught and replaced.                                                                                                                                                  |
| prompt_8 benchmark — which encoder / tagger?                               | Encoder: <b>Uni-Sign CSL→WLASL + Thai fine-tune</b> wins (val 0.5005) over CSL-only (0.442), scratch (0.342), Thai self-supervised (0.481), signer augmentation (0.4995), extractor-adversarial (0.4954). Tagger: <b>BiGRU</b> (selection 2.709) over TCN (2.642) and Transformer (2.541); self-training on 8.7 h of broadcasts did not help (2.659) and was not used. How2Sign / WLASL data were not added (§7).                                                                                                              |
| Does the model learn TSL or lean on the ASL/CSL prior?                     | It learns TSL: Thai fingerspelling (handshapes that do not exist in ASL/CSL) improves as much as lexical signs (0.82 → <b>0.90</b> vs 0.78 → 0.88); the representation moved substantially (CKA 0.60); removing the ASL step costs 0.06 on validation; training from scratch costs 0.16. But <b>38 %</b> of the remaining errors are confusions the zero-shot ASL/CSL model already made — the prior's confusions are only partly unlearned.                                                                                   |
| Cost                                                                       | GPU ≈ <b>\$12.8</b> (pose extraction on 4 Spot T4 \$3.1 · 8 encoder runs on Spot A100 \$8.6 · sequence stage \$1.1) + OpenAI ≈ \$1 + storage < \$1. After the round: GCS bucket emptied (kept), all 63 Vertex jobs deleted, no VM / disk / IP / endpoint exists.                                                                                                                                                                                                                                                               |

## 1. What changed from v5, and why (prompt\_7 bottlenecks)

| v5 bottleneck (prompt_7)           | v6 action                                                                                                                | result                                                                                                                                                                                                                                                        |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (1) bank depth decides recognition | th_sl added (7,719 clips, 8 signer clusters); ไหม checked in every source; per-depth calibration tried                   | multi-signer concepts 605 → 1,969; ร้องไห้ / ดื่มน้ำ / น้ำ now found on data_test. ไหม is in <b>no</b> source (only ไหม, ไหมพรม). Depth as a ranking prior or calibration feature was <b>rejected</b> — it encodes which dataset recorded most signers (§5.3) |
| (2) RTMW ↔ MediaPipe extractor gap | every video re-extracted with the inference extractor; extractor-adversarial loss tried                                  | adversarial loss did not help (val 0.4954 < 0.5005); the gap remains on the single TSL51 test signer (MP 0.620 vs RTMW 0.342 open-vocab), but RTMW sources now carry most of the vocabulary                                                                   |
| (3) accepted precision 0.76        | calibration re-fitted on real sentences, thresholds for precision ≥ 0.85 on tune, priors kept out of segmentation        | <b>0.85</b> (RTMW) / <b>0.91</b> (MP) on held-out templates; data_test 7/7                                                                                                                                                                                    |
| (4) tagger boundaries on real data | tagger benchmark (BiGRU/TCN/Transformer), real speed-perturbed sentences, self-training on 8.7 h of unlabeled broadcasts | BiGRU kept; self-training did not improve held-out real templates → not used; boundaries vs pseudo-labels remain weak (§5.1)                                                                                                                                  |
| (5) hubness                        | CSLS hubness correction implemented and tuned                                                                            | helps RTMW dictionary queries (th_sl val +0.014 MRR, TTRS val +0.053) but hurts conversational sentences → off in the final decoder (§5.3)                                                                                                                    |
| (6) fast phone signing             | tempo normalisation (time-stretch to training pace)                                                                      | 2× speed exact count 27 % → <b>67 %</b> , recall@1 0.38 → 0.46                                                                                                                                                                                                |

## 2. Data (details: [architecture\\_v6.md](#) §2–3, [data\\_prep/SCHEMA.md](#))



|                                                      | v5                      | v6                                                                                    |
|------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------|
| clips / videos in the standard schema                | 12,131                  | <b>19,897</b> (+ 102 continuous videos)                                               |
| hours of pose                                        | —                       | <b>26.7 h</b> (22.7 h RTMW)                                                           |
| isolated clips · concepts                            | 11,542 · 4,570          | <b>19,211 · 9,869</b>                                                                 |
| concepts with $\geq 2$ / $\geq 3$ / $\geq 4$ signers | 605 / – / –             | <b>1,969 / 567 / 280</b>                                                              |
| signer ids                                           | –                       | 151 ( $\approx$ 50 identifiable people + $\approx$ 80 YouTube signers with 1–3 clips) |
| TTRS poses                                           | v4, 12.5 fps up-sampled | <b>re-extracted, 25 fps, same code as inference</b>                                   |

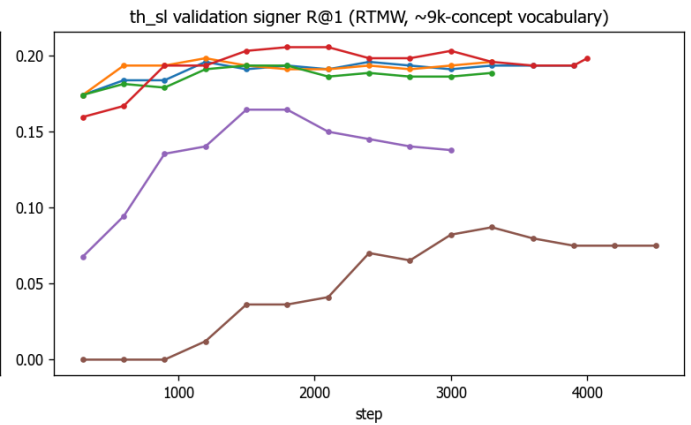
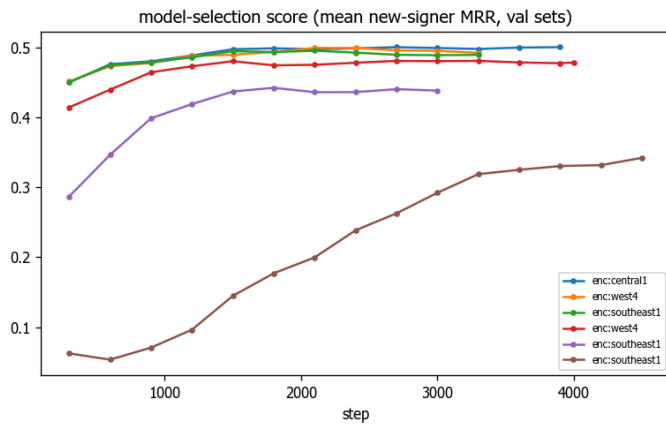
Held-out roles: v5 roles kept for every v5 clip (identical test sets); th\_sl val / test = one visually pure signer cluster each (1,203 / 1,338 clips), checked against 340 TTRS / TSL51 faces for identity leakage (none).

### 3. Benchmarks (prompt\_8)

#### 3.1 Encoder (same data, losses, schedule; selection = mean new-signer MRR on TTRS / TSL-ONE-S / th\_sl val signers)

| encoder                                            | val (select)                                     | TTRS test    | TSL-ONE-S test | th_sl test   | TSL51 RTMW | TSL51 MP     | YouTube      | GPU min |
|----------------------------------------------------|--------------------------------------------------|--------------|----------------|--------------|------------|--------------|--------------|---------|
| <b>Uni-Sign CSL→WLASL + Thai (all data)</b>        | <b>0.5005</b>                                    | <b>0.190</b> | 0.890          | 0.105        | 0.342      | 0.620        | 0.235        | 43      |
| + signer augmentation (retarget + mirror)          | 0.4995                                           | 0.167        | <b>0.896</b>   | 0.112        | 0.330      | <b>0.655</b> | 0.235        | 42      |
| + extractor-adversarial (GRL)                      | 0.4954                                           | 0.190        | 0.883          | <b>0.116</b> | 0.340      | 0.601        | <b>0.294</b> | 32      |
| Thai self-supervised (1,500 SSL steps) → fine-tune | 0.4809                                           | 0.179        | 0.885          | 0.112        | 0.281      | 0.622        | 0.235        | 73      |
| Uni-Sign CSL only (no ASL step)                    | 0.4424                                           | 0.107        | 0.814          | 0.090        | 0.218      | 0.608        | 0.118        | 30      |
| same architecture from scratch                     | 0.3422 (stopped at 4,500 steps, still far below) | –            | –              | –            | –          | –            | –            | 45      |
| data ablation: without th_sl                       | 0.4676                                           | 0.131        | 0.858          | 0.116        | 0.389      | 0.676        | 0.235        | 19      |
| data ablation: without MediaPipe sources           | 0.3538                                           | 0.179        | 0.780          | 0.101        | 0.361      | 0.618        | 0.176        | 47      |

All test columns use **the same bank** (every original training clip) and vocabulary, so only the encoder differs. Choice by validation only: the all-data encoder. Reading: pretraining is essential (scratch), the ASL step helps (CSL-only), the Thai self-supervised stage did not add to it (its contrastive task stayed too easy — view-match accuracy 1.00 — while masked-pose loss fell), augmentation and the adversarial loss are within noise, and th\_sl helps TTRS / TSL-ONE-S but costs ~5 points on the TSL51 researcher (\$4).



### 3.2 Tagger (held-out real *tagger-val* templates; sentence-test templates never used)

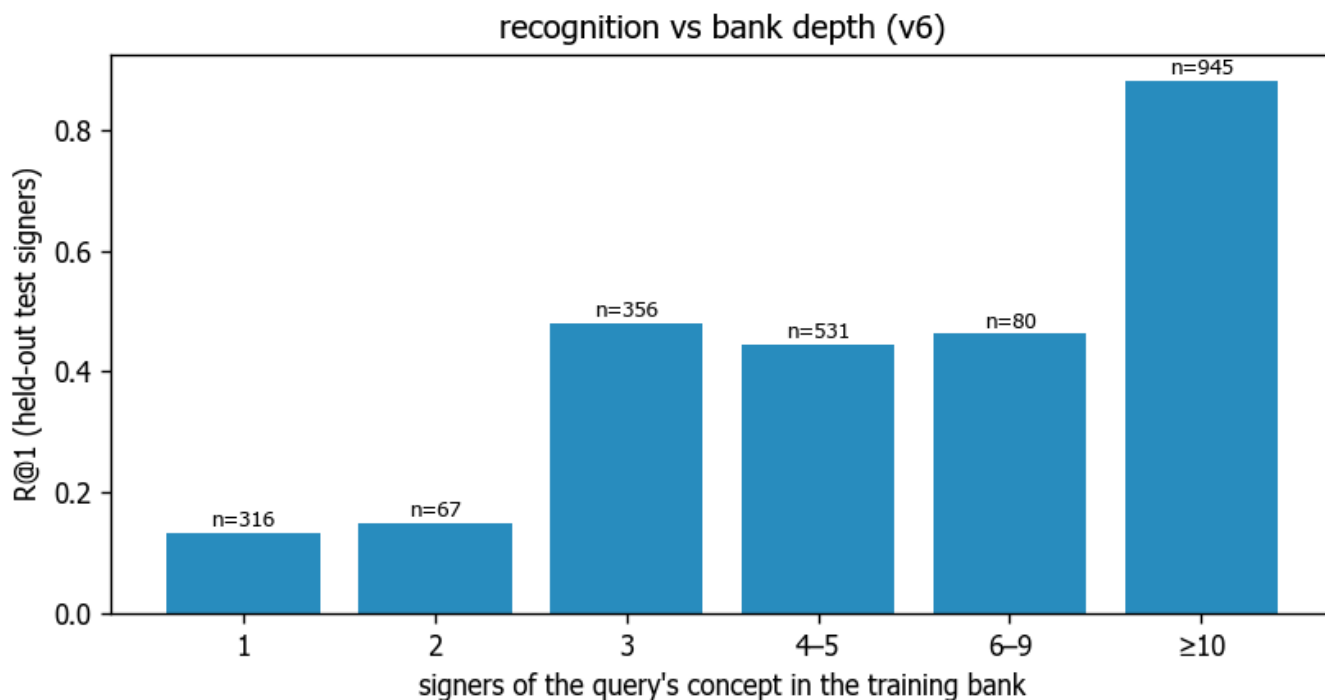
| tagger                                                                        | selection    | boundary F1 | sign-frame F1 | count error |
|-------------------------------------------------------------------------------|--------------|-------------|---------------|-------------|
| <b>BiGRU</b>                                                                  | <b>2.709</b> | 0.202       | 0.694         | 0.52        |
| TCN (dilated)                                                                 | 2.642        | 0.172       | 0.673         | 0.64        |
| Transformer encoder                                                           | 2.541        | 0.142       | 0.705         | 0.79        |
| BiGRU + self-training (1,209 pseudo-labelled 24 s chunks from 102 broadcasts) | 2.659        | —           | —             | —           |

## 4. Isolated signs, held-out signers ([artifacts/reports/isolated.json](#), [baseline\\_v5.json](#))

Same queries, bank = training clips of v6 (open vocabulary  $\approx$  9.9k concepts):

| protocol                     | n   | v5 encoder   | v6 encoder           | zero-shot Uni-Sign |
|------------------------------|-----|--------------|----------------------|--------------------|
| TTRS test signers            | 84  | 0.155        | <b>0.190</b>         | 0.119              |
| TSL-ONE-S 5 unseen signers   | 873 | 0.864        | <b>0.890</b>         | 0.796              |
| th_sl unseen signer (RTMW)   | 277 | 0.090        | <b>0.105</b>         | 0.083              |
| YouTube title signers        | 17  | 0.176        | <b>0.235</b>         | 0.294              |
| TSL51 researcher, MediaPipe  | 521 | <b>0.633</b> | 0.620                | 0.608              |
| TSL51 researcher, RTMW       | 524 | <b>0.368</b> | 0.342                | 0.372              |
| null vs sign AUC (RTMW / MP) | 547 | —            | <b>0.951 / 0.895</b> | 0.861 / 0.839      |

v5 *as shipped* (its own 4.5k-concept bank) scores TSL51-RTMW 0.525 — the drop to  $\sim$ 0.35 is mostly the **vocabulary doubling** (th\_sl adds  $\sim$ 5k concepts, many visually close), not a worse encoder. Recognition by bank depth (held-out test signers):



$R@1 \approx 0.13\text{--}0.15$  for words with 1–2 training signers,  $\approx 0.45\text{--}0.48$  with 3–9, 0.88 with  $\geq 10$ . **Signers per word remain the strongest lever** (prompt\_7's diagnosis confirmed with v6 data).

A zero-shot + fine-tuned score fusion would raise TSL51-RTMW (0.488  $\rightarrow$  0.547 MRR) but lowers the validation objective (0.5003  $\rightarrow$  0.4933)  $\rightarrow$  not used (choosing it from test numbers would be tuning on test).

## 5. Continuous signing (held-out TSL51 templates; [artifacts/reports/spotting.json](#))

### 5.1 Final decoder vs v5

| test set            | seg F1       | exact count  | count err   | recall@1     | recall@5     | prec@1       | WER                     |
|---------------------|--------------|--------------|-------------|--------------|--------------|--------------|-------------------------|
| v5 RTMW             | <b>0.835</b> | <b>0.71</b>  | <b>0.35</b> | 0.473        | <b>0.718</b> | <b>0.486</b> | 0.524 (re-ranked 0.467) |
| <b>v6 RTMW</b>      | 0.793        | 0.635        | 0.49        | <b>0.487</b> | 0.651        | 0.474        | 0.571 (re-ranked 0.588) |
| v5 RTMW 1.5×        | <b>0.817</b> | 0.54         | 0.56        | 0.456        | <b>0.662</b> | <b>0.524</b> | <b>0.530</b>            |
| <b>v6 RTMW 1.5×</b> | 0.800        | <b>0.683</b> | <b>0.43</b> | <b>0.467</b> | 0.647        | 0.467        | 0.570                   |
| v5 RTMW 2×          | 0.723        | 0.27         | 0.92        | 0.382        | 0.542        | 0.509        | 0.597                   |
| <b>v6 RTMW 2×</b>   | <b>0.786</b> | <b>0.667</b> | <b>0.42</b> | <b>0.456</b> | <b>0.629</b> | 0.478        | <b>0.558</b>            |
| v5 MediaPipe        | <b>0.851</b> | 0.71         | 0.33        | <b>0.653</b> | <b>0.776</b> | <b>0.689</b> | 0.333 (0.320)           |
| <b>v6 MediaPipe</b> | 0.839        | <b>0.730</b> | 0.41        | 0.600        | 0.767        | 0.566        | 0.442 (0.425)           |

**Status tiers (held-out templates)** — accepted words: RTMW **0.85 precision** (27 % of segments; v5 0.76 at 50 %), MediaPipe **0.91** (46 %; v5 0.84 at 70 %), RTMW 2× **0.83** (v5 0.77). v6 accepts fewer words but they are right more often — the behaviour data\_test shows (7/7).

## 5.2 Components on the test templates (each switched on in turn)

|                                                | RTMW seg F1 / exact / R@1 / WER             | 2× seg F1 / exact / R@1 / WER                             |
|------------------------------------------------|---------------------------------------------|-----------------------------------------------------------|
| decoder only                                   | 0.793 / 0.635 / 0.411 / 0.651               | 0.690 / 0.310 / 0.340 / 0.633                             |
| + Thai word-frequency prior (recognition only) | 0.793 / 0.635 / <b>0.487</b> / <b>0.571</b> | 0.690 / 0.310 / 0.398 / 0.553                             |
| + tempo normalisation (final)                  | 0.793 / 0.635 / 0.487 / 0.571               | <b>0.786</b> / <b>0.667</b> / <b>0.456</b> / <b>0.558</b> |

## 5.3 Decisions, including the ones that were reversed (all on held-out validation data; data\_test only reported)

1. **Priors must not touch segmentation.** Adding word priors to window scores inflated window gains and over-split signs (exact count 0.64 → 0.38); segmentation now uses visual scores only and priors re-rank words inside each segment.
2. **Tune templates are not a valid selection set** — the tagger trained on them (segment F1 ≈ 1.0 there); decoder priors and tempo are chosen on the held-out tagger-val templates.
3. **Bank depth was rejected as a ranking prior and as a calibration feature.** It improved TSL51 validation strongly (0.774 → 0.894) but the mechanism is a dataset confound: TSL51's 51 words overlap TSL-ONE-S's family / number vocabulary, which has 29 signers. On real video it surfaced TSL-ONE-S words (ศูนย์, หก, แม่, เด็ก, ประเทศไทย) everywhere. Removed on that principle; reported as runs A/B in §6.
4. **CSLS hubness correction:** on broad dictionary validation (th\_sl / TTRS / TSL-ONE-S val signers)  $\alpha = 0.5 + \gamma = 0.1$  is best (MRR 0.5005 → 0.5163), but on conversational sentences it penalises exactly the common words a conversation uses. The deployment target is conversation → selection on the conversational validation set →  **$\alpha = 0$ ,  $\gamma = 0.25$** .

## 6. Real-world: data\_test (from the raw videos; [artifacts/reports/infer\\_final.json](#), [result\\_reporting/inference/](#))

| run (decided on held-out data, data_test only reported)                | top-1 in some segment | top-5        | accepted correct |
|------------------------------------------------------------------------|-----------------------|--------------|------------------|
| v5 final (run B)                                                       | 7/22                  | 10/22        | 6/12             |
| v6 run A: depth prior + depth calibration                              | 7/22                  | 14/22        | 5/7              |
| v6 run B: frequency prior + depth calibration                          | 7/22                  | 14/22        | 7/17             |
| v6 run C: broad-val CSLS + frequency prior                             | 7/22                  | 10/22        | 2/2              |
| <b>v6 final: conversational-val frequency prior, no depth anywhere</b> | <b>9/22</b>           | <b>14/22</b> | <b>7/7</b>       |

Per video (final; word accepted · rank of the reference word among a segment's top-5):

| video                        | reference                                    | segments | top-1 per segment                                                            | accepted                | reference ranks                                                       | sentence shown                                    |
|------------------------------|----------------------------------------------|----------|------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------|---------------------------------------------------|
| test1                        | สวัสดี สบายดี ชอบ<br>โกธร แมว แพน ปรบ<br>มือ | 7 / 7    | สวัสดี · สบายดี · ฉัน · อะ<br>โดบี อิลลัสเตรเตอร์ · แมว ·<br>เหมือนกัน · แมว | สวัสดี, สบาย<br>ดี, แมว | สวัสดี 1 · สบายดี 1 · แมว<br>1 · โกธร 2 · ชอบ 4 · แพน<br>– · ปรบมือ – | "สวัสดี ฉันสบายดี<br>แมวเหมือนกันไหม"<br>(partly) |
| ฉันปलอบ<br>เพื่อน<br>ร้องไห้ | ฉัน ปลอบใจ เพื่อน<br>ร้องไห้                 | 4 / 4    | เพื่อน · ร้องไห้ · สงสาร ·<br>เด็ก                                           | เพื่อน                  | เพื่อน 1 · ร้องไห้ 1 · ปลอบ<br>ใจ (สงสาร/ปลอบโยน<br>nearby) –         | "เพื่อนร้องไห้สงสาร<br>เด็ก" (partly)             |
| ฉันรัก<br>เพื่อน             | ฉัน รัก เพื่อน                               | 2 / 3    | แต่งงาน · เด็ก                                                               | –                       | ฉัน 2                                                                 | withheld                                          |
| พ่อต๋มน้ำ                    | พ่อ ต๋ม น้ำ                                  | 6 / 3    | ศูนย์ · พ่อ · ตา · ต๋ม · เด็ก<br>· เด็ก                                      | พ่อ                     | พ่อ 1 · ต๋ม 1 · น้ำ 2                                                 | "พ่อ ตา ต๋ม เด็ก"<br>(wrong extra<br>words)       |
| ไปทานข้าว<br>ด้วยกันมั๊ย     | ไป กิน ข้าว ด้วยกัน<br>ไหม                   | 4 / 5    | ฉัน · เดียวกัน · กิน · ไป                                                    | กิน, ไป                 | ไป 1 · กิน 1 · ข้าว 5 · ไหม<br>not in any data                        | "ฉันกินเหมือนกันไป"<br>(partly)                   |

Remaining real-world errors, honestly: 3-signs-in-1.9 s phone signing (ฉันรักเพื่อน) still becomes 2 segments; extra segments on hands moving between signs in พ่อต๋มน้ำ (6 for 3) are output as uncertain words (เด็ก?) that the sentence layer then uses; รัก / ปรบมือ / แพน / ปลอบใจ are not found (1–3 training signers); ไหม cannot be found (no source has it).

## 7. What was not done, and why

- **How2Sign / WLASL as training data (prompt\_8):** How2Sign has sentence-level alignment only (no per-sign timing), so it would add unlabeled continuous signing — which v6 already tested with 8.7 h of Thai broadcasts (self-training: no gain) — plus the ASL-confusion risk you named. Not downloaded (large external download; it would need your OK).
- **Re-training the tagger on all tune templates** (v5 trained on all; v6 held ¼ out for honest selection, which likely explains part of the natural-pace count gap): one more A100 job (~\$1.1). Not run under the cost limit of this round; it is the cheapest next step for \$5.1.

## 8. Cost and GPU usage

| job group                                                                      | GPU       | minutes    | ≈ USD         |
|--------------------------------------------------------------------------------|-----------|------------|---------------|
| pose extraction (13,663 videos, 2.0 M frames; 4 shards + retries, smoke tests) | Spot T4   | 486        | 3.08          |
| encoder runs (all, adv, nothsl, nomp, csl, scratch, ssl, aug)                  | Spot A100 | 334        | 8.62          |
| sequence stage (alignment, synthesis, tagger benchmark, self-training)         | Spot A100 | 43         | 1.10          |
| <b>total GPU</b>                                                               |           | <b>863</b> | <b>≈ 12.8</b> |

Local (free): face embeddings / signer clusters, ingest, lexicon, speed + tempo tables (2 h on the laptop GPU), tuning, evaluation, data\_test, ONNX export. Lessons: one-session GPU memory caps (an ORT arena OOM made one worker per shard fail every clip), a stall watchdog, Cloud Storage client staging instead of the bucket mount, cancel benchmark losers early.

**Clean-up done:** gs://temp-gpu-bucket-job emptied (0 objects, bucket kept, soft-delete retention 0 → nothing billed), all 63 Vertex AI custom jobs deleted in 6 regions, no Compute Engine instance / disk / address, no Vertex endpoint / model.

## 9. Reproduce

---

```
python main.py data pull data_prep # or rebuild: python main.py prep ... (README)
python main.py cloud upload-code --tag v6 && python main.py cloud upload-data && python main.py cloud upload-models
python main.py cloud launch --tag v6 --stage train_encoder --args "--steps 4000 --eval_every 300 --patience 4 --P 48 --K 2 --ema 0.998"
python main.py cloud launch --tag v6 --stage sequence --args "--encoder_run <encoder_run> --epochs 20 --n_synth 3000"
python main.py eval build --run <encoder_run> --seq <sequence_run> && python main.py eval grammar
python main.py eval fast --seq <sequence_run> && python main.py eval spot --seq <sequence_run> --no_depth_calibration
python main.py eval isolated --run <encoder_run> && python main.py eval baseline --run <encoder_run>
python main.py test && python main.py eval vocab && python main.py export-onnx
```

Runs used: encoder `enc-all-0241-us-central1`, sequence `seq-all-0340-asia-southeast1`  
(`models/config.json`).